

Does Volatility Persistence Forecast Volatility? Rolling Long Memory, Crisis Windows, and Implied Volatility

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Abstract

Rolling estimates of long memory in volatility rise sharply in crises and are often read as a time-varying state of the market. We ask whether this state forecasts stock volatility beyond implied volatility, and what it measures. For 115 S&P 500 stocks over 2001–2026, we forecast log Parkinson variance one day, one week and one month ahead with per-stock HAR models, HAR models augmented with the VIX and MOVE indices (HAR-X), and specifications that add own-stock, sector and cross-sectional persistence. All forecasts are point-in-time, each design was fixed before its results were computed, and inference combines panel Diebold–Mariano, Clark–West and Giacomini–White tests. Persistence never improves significantly on HAR-X: the largest gain, 0.26% at one week, is not significant, and the full model is significantly worse at one day and one month. Pooled estimation, a term-structure target, a market-level series and five machine-learning estimators lead to the same conclusion. The persistence state itself is largely mechanical: it jumps when an extreme episode enters the estimation window and falls when the episode leaves, at dates that move one-for-one with the window length, and the largest VIX inside the window explains 83% of its variation. A one-day misalignment between HAR inputs and the VIX inflates the apparent forecasting gain from the VIX by about a third.

Keywords: long memory; volatility forecasting; implied volatility; HAR model; rolling-window estimation; spurious long memory; forecast evaluation

1. Introduction

Volatility is persistent. Shocks to the variance of stock returns decay slowly, the autocorrelations of squared and absolute returns stay positive for months, and the periodogram of realized variance rises steeply toward zero frequency (Andersen et al., 2003; Ding et al., 1993). A large literature models this with fractional integration of order d , and a sizeable applied literature estimates d , or the related Hurst exponent, over rolling windows and reads the resulting series as a time-varying property of the market: persistence that strengthens in crises, efficiency that improves or deteriorates, or adaptive behavior in the sense of Lo (2004). Examples range from early work on the Hurst exponent over time (Cajueiro & Tabak, 2004) to recent studies of volatility persistence after COVID-19 (Vera-Valdés, 2022) and of the pandemic's effect on market efficiency (Takaishi, 2025).

If rolling persistence is a genuine state of the market, it should carry information about future volatility. We test this implication and ask two questions. First, does a persistence state built from rolling long-memory estimates forecast stock volatility beyond what implied volatility already contains? Second, what does the rolling persistence state measure?

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We answer both on a panel of 115 S&P 500 stocks from November 2001 to April 2026. Volatility is the daily Parkinson range variance, and the forecast target is the log of its mean over the next 1, 5 or 22 trading days. The benchmarks are the heterogeneous autoregressive (HAR) model of [Corsi \(2009\)](#) and a HAR model augmented with the closing levels of the VIX and MOVE indices (HAR-X). The persistence state enters through own-stock rolling estimates of d , sector and cross-sectional averages of those estimates, and their interactions with implied volatility. Every forecast is point-in-time: predictors use only information available at the close of the forecast origin, and a training row enters a regression only if its target window has closed by the origin. Each specification and its decision rule were recorded in a dated experiment ledger before its results were computed. Inference combines the panel Diebold–Mariano test with the [Harvey et al. \(1997\)](#) correction, the [Clark and West \(2007\)](#) test for nested models and the conditional predictive ability test of [Giacomini and White \(2006\)](#).

The answer to the first question is negative. Implied volatility lowers the mean squared error of HAR by 3.1%, 5.5% and 1.8% at the three horizons, the last not significantly. Persistence does not improve significantly on HAR-X. Its largest gain is 0.26% at one week, with a Diebold–Mariano statistic of 0.41, and the full model is significantly worse than HAR-X at one day and one month. Pooled estimation with stock fixed effects, a target that measures the slope of the variance term structure, a single market-level series and five machine-learning estimators lead to the same conclusion. At the daily and weekly horizons the Clark–West test rejects the null that the persistence terms have zero population coefficients, but the Giacomini–White test never finds that the estimated models forecast better: the information exists in population but is too small to be exploited once the models are estimated.

The answer to the second question explains the first. The cross-sectional mean of rolling d estimates behaves as an indicator of whether a large and long volatility episode lies inside the estimation window. It jumps when such an episode enters the window and falls when the episode leaves, and the exit date moves one-for-one with the window length. For windows of 500, 750 and 1000 trading days the 2020 plateau ends on 11 March 2022, 10 March 2023 and 8 March 2024, each time with a fall in the bottom 0.2% of all four-week changes in the series. A short-memory ARMA model fitted to calm-period data, with the observed 2020 episode added, reproduces the calm level of the estimate, the plateau, the exit date and between half and two-thirds of the rise. The largest VIX inside the window explains 81–83% of the variation in the state, against 18–28% for the current VIX. A lagged summary of past extreme episodes has little to add to an implied-volatility index that already reflects them.

A third result is practical. If the HAR inputs end one day before the forecast origin while the VIX is the origin-day close, HAR is at an informational disadvantage and the gain from the VIX is overstated by about a third. With aligned inputs the error of HAR falls by 8.1% at the daily horizon, and the gain of HAR-X over HAR falls from 5.1% to 3.1%.

These results connect to three literatures. The first shows that structural breaks, level shifts and regime switching produce spurious evidence of long memory ([Diebold & Inoue, 2001](#); [Granger & Hyung, 2004](#); [Mikosch & Stărică, 2004](#); [Perron & Qu, 2010](#); [Smith, 2005](#)). It has developed tests and estimators that separate the two ([Hou & Perron, 2014](#); [Qu, 2011](#)) and finds that little genuine long memory survives once level shifts are modeled ([Varneskov & Perron, 2017](#)). That literature concerns full-sample estimates. We document its consequence for rolling estimates, which are what applied work interprets, and propose a simple check: vary the window and see whether the turning points move with it. The problem has a long history in other sciences. Mean shifts were shown to mimic

the Hurst phenomenon in river flows (Klemeš, 1974), and the distinction between extreme events and persistence goes back to the “Noah” and “Joseph” effects of Mandelbrot and Wallis (1968). Patchy composition mimics long-range correlation in DNA (Karlin & Brendel, 1993), and rolling-window indicators of critical slowing down and of dynamic brain connectivity have faced the same critique (Boettiger & Hastings, 2012; Dakos et al., 2008; Ditlevsen & Johnsen, 2010; Leonardi & Van De Ville, 2015).

The second literature compares implied and time-series forecasts of volatility (Blair et al., 2001; Busch et al., 2011; Christensen & Prabhala, 1998; Fleming, 1998) and documents a variance risk premium that varies with market conditions (Andersen et al., 2015; Bates, 2000; Bekaert & Hoerova, 2014; Chernov, 2007). Our timing result identifies a concrete pitfall in that comparison. The third literature asks whether volatility is rough (Cont & Das, 2024; Fukasawa et al., 2022; Gatheral et al., 2018) and whether volatility forecasts create value in managed portfolios (Barroso & Detzel, 2021; Cederburg et al., 2020; Moreira & Muir, 2017). We address both briefly. A roughness estimate of 0.06 on daily range data is reproduced by a model without roughness once the sampling noise of the range is added, and managed portfolios built on any of our forecasts have the same Sharpe ratio as the equal-weighted portfolio when returns are measured in excess of the Treasury bill rate.

The paper proceeds as follows. Section 2 describes the data and the full-sample memory estimates. Section 3 shows what the rolling persistence state measures. Section 4 sets out the forecasting design, and Section 5 reports the results. Section 6 presents robustness checks and the portfolio exercise, and Section 7 discusses implications and limitations. The appendices contain supplementary results and the registration record.

2. Data and Measurement

2.1. Sample

Daily open, high, low and close prices for S&P 500 constituents were obtained from Bloomberg. The panel retains the 115 stocks observed on at least 70% of the 6136 trading days between 29 November 2001 and 21 April 2026; it covers all eleven GICS sectors. Membership is that of the April 2026 index, so firms that left the index earlier are excluded; Section 7 discusses the implications. Returns are daily log price returns. For use as predictors they are winsorized point-in-time at the expanding 0.1% and 99.9% quantiles after a 500-day warm-up, so that no observation is trimmed with information from later dates. One special dividend that appears as a price drop (Keurig Dr Pepper, 10 July 2018) is added back. The market variables are the daily closes of the Cboe VIX index and of the ICE BofA MOVE index of Treasury implied volatility; the three-month Treasury bill rate is used in the portfolio exercise.

2.2. Variance

Daily variance is measured by the range estimator of Parkinson (1980),

$$\sigma_{i,t}^2 = \frac{(\ln H_{i,t} - \ln L_{i,t})^2}{4 \ln 2}, \quad (1)$$

where $H_{i,t}$ and $L_{i,t}$ are the day's high and low prices. The estimator uses intraday information, is several times more efficient than the squared close-to-close return, and is available for the whole sample, which five-minute realized variance is not. Days with a zero range are treated as missing. All models use the logarithm of variance, which is close to Gaussian and makes forecast errors comparable across stocks.

2.3. Timing

The forecast origin is the close of trading day t , and every predictor uses only information available at that time, including the day's range and return and the closing levels of the VIX and MOVE indices. The VIX is computed until 4:15 p.m. Eastern time, fifteen minutes after the equity close; we treat the two as simultaneous. Section 5 shows that this alignment matters. The forecast target for horizon h is

$$y_{i,t,h} = \ln\left(\frac{1}{h} \sum_{k=1}^h \sigma_{i,t+k}^2\right), \quad h \in \{1, 5, 22\}, \quad (2)$$

the log of mean variance over the next day, week and month. Forecasts are made every fifth trading day, which gives 1078 weekly origins from 22 November 2004, the first date at which a 750-day window is available, to 21 April 2026.

2.4. Full-sample memory

Table 1 reports full-sample estimates of the fractional parameter d obtained with the log-periodogram estimator of Geweke and Porter-Hudak (1983) (GPH) and the local Whittle estimator of Robinson (1995), both with bandwidth $m = T^{0.65}$. Returns show no evidence of long memory: the mean estimate is -0.02 . Log variance is strongly persistent: the mean GPH estimate is 0.54 , the mean local Whittle estimate is 0.52 , and 84% of stocks have a GPH estimate above 0.5 , the boundary of stationarity. These values are in line with the established stylized facts (Andersen et al., 2003). A Hurst exponent of 0.06 , obtained from the scaling of increments of log variance, would indicate volatility much rougher than Brownian motion (Gatheral et al., 2018). Appendix A shows that a two-factor Markovian model without roughness, observed through the sampling noise of the range estimator, produces the same value, so roughness is not identified at the daily frequency (Cont & Das, 2024). We do not use roughness further.

Table 1. Full-sample memory estimates across the 115 stocks.

Series	Estimator	Mean	10th pct	Median	90th pct	Share > 0.5
Returns	GPH	-0.020	-0.082	-0.022	0.044	0%
	LW	-0.027	-0.077	-0.027	0.025	0%
Log Parkinson variance	GPH	0.538	0.490	0.537	0.587	84%
	LW	0.516	0.476	0.515	0.559	70%
Log Parkinson variance	Hurst ($q = 2$)	0.060	0.053	0.060	0.069	—

Notes: Daily data, 29 November 2001 to 21 April 2026 (6136 trading days). GPH: log-periodogram estimator of the fractional parameter d ; LW: local Whittle; bandwidth $m = T^{0.65}$. Hurst: scaling of second moments of increments of log variance over lags 1 to 21 days. Percentiles are across stocks.

Estimates of d above 0.5 for most stocks call for caution, since they imply nonstationary log variance. Long memory and a slowly decaying short-memory process with occasional large episodes are difficult to distinguish in samples of this length (Diebold & Inoue, 2001; Granger & Hyung, 2004). The rolling estimates in the next section make the difference visible.

3. What the Rolling Persistence State Measures

3.1. The state

For each stock and weekly origin, $\hat{d}_{i,t}(W)$ denotes the GPH estimate computed from the W log-variance observations ending on day t , with $W = 750$ in the baseline. The persistence state is the cross-sectional mean $\bar{d}_t(W) = N^{-1} \sum_i \hat{d}_{i,t}(W)$, together with its cross-sectional standard deviation. The state rises sharply in crises: it averages about 0.38

in calm years and 0.57 while the 2020 episode is inside the window. Read as a property of the market, this would mean that shocks to volatility became markedly more persistent after March 2020 and remained so for three years.

3.2. *A mechanical account*

A simpler account predicts the same rise and makes three further predictions that the persistence interpretation does not. Suppose that log variance is a short-memory process plus a temporary episode, a block of elevated values of height b and length L . Within a window of length W , the episode adds a component to the discrete Fourier transform whose modulus at the Fourier frequency $\lambda_j = 2\pi j/W$ is $b |\sin(L\lambda_j/2)/\sin(\lambda_j/2)|$. This modulus is large at frequencies below about $2\pi/L$ and does not depend on the position of the episode within the window, because shifting the episode changes only the phase. The log-periodogram estimator regresses the log periodogram on log frequency over the lowest m frequencies, so the additional low-frequency power raises $\hat{d}$. Three predictions follow. First, the estimate jumps when the episode enters the window, stays roughly constant while the episode is inside, and falls when it leaves; the fall therefore occurs W days after the episode and moves one-for-one with the window length. Second, the size of the effect increases with the duration of the episode, not only with its peak, so a one-day crash should have little effect. Third, the state should track the largest recent episode more closely than the current level of volatility. None of these predictions requires the persistence of shocks to change.

3.3. *Evidence*

The events are fixed in advance as the three largest days of cross-sectional mean log variance that lie at least 250 trading days apart: 10 October 2008, 6 May 2010 (the Flash Crash) and 18 March 2020. For each event and window length $W \in \{500, 750, 1000\}$, Panel A of Table 2 compares the mean state over the four weeks after the event day enters the window with the mean over the four weeks before, and does the same when the day leaves the window W trading days later. Each change is ranked among all four-week changes in the series.

Table 2. What the rolling persistence state tracks.

Panel A. Change in $\bar{d}_t(W)$ when the event day enters and leaves the window (percentile among all changes)						
Event	W	Entry	Pct	Exit date	Exit	Pct
2008-10-10	500	+0.159	99.8	2010-10-06	+0.009	80.8
	750	+0.093	99.6	2011-10-03	-0.065	0.8
	1000	+0.104	99.5	2012-09-28	-0.036	1.7
2010-05-06	500	-0.008	22.7	2012-04-30	-0.000	51.6
	750	+0.012	90.4	2013-04-30	-0.017	9.6
	1000	+0.002	65.9	2014-04-28	-0.022	5.4
2020-03-18	500	+0.181	99.9	2022-03-11	-0.164	0.2
	750	+0.125	99.9	2023-03-10	-0.108	0.1
	1000	+0.108	99.7	2024-03-08	-0.090	0.1
Panel B. R^2 of $\bar{d}_t(W)$ on the VIX						
	W	Max VIX in window			Current VIX	
	500	0.81			0.28	
	750	0.83			0.27	
	1000	0.82			0.18	
Panel C. The 2020 episode: data versus a short-memory null						
	W	Before	Inside	After exit	Rise	Largest drop
Data	500	0.35	0.60	0.33	+0.25	
Null	500	0.41	0.54	0.47	+0.13	day 509
Data	750	0.38	0.57	0.33	+0.19	
Null	750	0.40	0.53	0.46	+0.13	day 759
Data	1000	0.37	0.53	0.34	+0.16	
Null	1000	0.39	0.50	0.44	+0.11	day 1009

Notes: $\bar{d}_t(W)$ is the cross-sectional mean of the GPH estimate on log Parkinson variance over the W trading days ending on day t , on the weekly grid. Events are the three largest days of cross-sectional mean log variance at least 250 days apart (rule fixed in advance). Entry and exit changes compare the four weekly estimates after the date with the four before; Pct is the percentile of that change among all such changes. Panel C: mean $\bar{d}_t$ in the 120 days before the event, from day 60 to day $W - 60$, and from day $W + 20$ to $W + 120$. The null is an ARMA(1,1) with the median parameters of per-stock fits to 2012–2019 log variance plus the observed one-year 2020 excess log variance, 200 replications; Largest drop is the day (relative to the event) of its largest four-week fall.

The 2020 event produces the predicted pattern at every window length. The state rises by 0.11 to 0.18 on entry, among the largest 0.3% of changes, and falls by 0.09 to 0.16 on exit, among the smallest 0.2%. The exit dates are 11 March 2022, 10 March 2023 and 8 March 2024. Figure 1 shows the three series: each holds a plateau and drops on its own exit date. The October 2008 event shows the same pattern at 750 and 1000 days. At 500 days its exit is masked because the remainder of the 2008–2009 episode and the May 2010 turmoil are still inside the window. The Flash Crash, a single extreme day, barely moves the state on entry or exit, as the second prediction requires.

Panel B tests the third prediction. The largest VIX close inside the same window explains 81–83% of the variation in the state at every window length, whereas the current VIX explains 18–28%. Most of the variation in the state is therefore accounted for by a single observable, the highest VIX close in the estimation window.

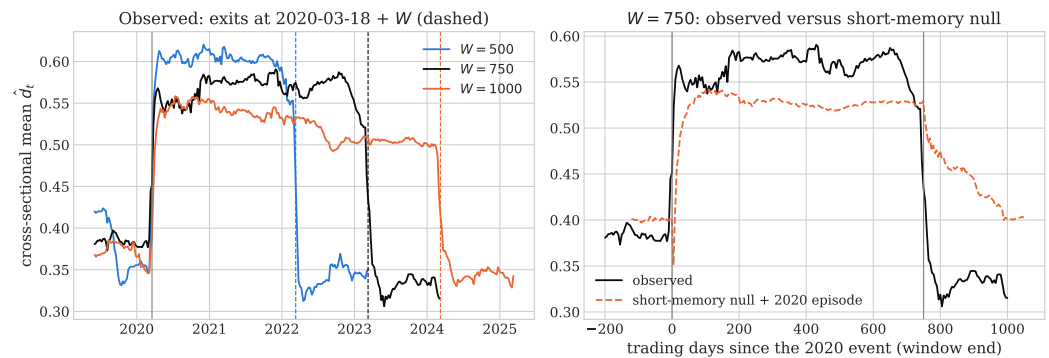


Figure 1. The persistence state around the 2020 episode. Left: cross-sectional mean of the rolling GPH estimate for windows of 500, 750 and 1000 trading days; the solid gray line marks 18 March 2020 and the dashed lines the dates on which that day leaves each window. Right: the 750-day series in event time and the mean of 200 simulations of a short-memory ARMA(1,1) process, fitted to 2012–2019 data, with the observed 2020 excess log variance added.

Panel C and the right panel of Figure 1 ask whether a process without long memory can generate this pattern. We fit an ARMA(1,1) model to each stock's log variance over 2012–2019, a period without an episode of the scale of 2008 or 2020, take the median parameters ($\phi = 0.91$, $\theta = -0.69$), add the observed cross-sectional median excess log variance of the year after 18 March 2020, and apply the same rolling estimator to 200 simulated paths. The simulated state starts at 0.40, close to the observed 0.385, rises on entry, holds a plateau, and has its largest fall within nine days of the exit date at every window length. It reproduces 53–69% of the observed rise. A shortfall is to be expected, since the null contains a single stylized episode with the cross-sectional median profile, whereas each stock's window in the data also contains its own later shocks of 2020–2022.

3.4. Implications

Two implications matter for what follows. First, a persistence state built from rolling long-memory estimates is a lagged and smoothed record of how much extreme volatility has occurred over the past few years. Implied volatility already reflects that history, partly through the variance risk premium, which rises after crises and decays slowly (Andersen et al., 2015; Bates, 2000). There is therefore little reason to expect the state to add forecasting information, and Sections 5 and 6 find that it does not.

Second, rolling memory estimates should not be read as evidence of changing persistence or efficiency without checking how their turning points depend on the window. The dates illustrate the risk. With a 750-day window the 2020 shock leaves the window on 10 March 2023, the day Silicon Valley Bank failed; with a 500-day window it leaves on 11 March 2022, two weeks after Russia invaded Ukraine. A study based on a single window could plausibly attribute the fall in persistence to one of those events. Varying the window and checking whether the turning points move with it is an inexpensive safeguard. Our evidence concerns memory in volatility; whether rolling estimates of memory in returns behave in the same way depends on the estimator and is left for future work.

4. Forecasting Design

4.1. Models

All models are linear regressions of the target (2) on predictors dated t , estimated stock by stock. The baseline, model A, is the HAR model of Corsi (2009) in logs with a return term,

$$y_{i,t,h} = \beta_0 + \beta_d \ln RV_{i,t}^{(1)} + \beta_w \ln RV_{i,t}^{(5)} + \beta_m \ln RV_{i,t}^{(22)} + \gamma_1 r_{i,t} + \gamma_2 |r_{i,t}| + \varepsilon_{i,t,h}, \quad (3)$$

where $RV_{i,t}^{(k)}$ is the mean Parkinson variance over days $t - k + 1$ to t , and $r_{i,t}$ is the return on day t . HAR-X, model A_1 , adds the levels of the VIX and MOVE indices at the close of day t . The persistence blocks are added to HAR one at a time and then jointly. Three further specifications, fixed before their results were computed, add a single block to HAR-X, so that the comparison with HAR-X isolates that block. Table 3 lists the predictors.

Table 3. Model specifications.

Model	Predictors
A (HAR)	$\ln RV^{(1)}, \ln RV^{(5)}, \ln RV^{(22)}, r_t, r_t $
A_1 (HAR-X)	HAR + VIX, MOVE
A_2	HAR + own-stock $\hat{d}_{i,t}$, its weekly change, its standard deviation and slope over four weeks, own Hurst exponent and its weekly change
A_3	HAR + cross-sectional mean $\bar{d}_t$ and dispersion $\sigma_{d,t}$
A_4	HAR + leave-one-out sector mean of $\hat{d}_{i,t}$
A_5	HAR + $\hat{d}_{i,t}$, VIX, MOVE, $\hat{d}_{i,t} \times \text{VIX}$, $\hat{d}_{i,t} \times \text{MOVE}$
C	Union of A_1 to A_5 (18 predictors)
<i>Registered HAR-X extensions</i>	
HAR-X + cross-sectional state	HAR-X + $\bar{d}_t, \sigma_{d,t}$
HAR-X + sector state	HAR-X + leave-one-out sector mean of $\hat{d}_{i,t}$
HAR-X + state $\times$ HAR terms	HAR-X + $\bar{d}_t$ and $\bar{d}_t \times \ln RV_{i,t}^{(k)}, k = 1, 5, 22$

Notes: All predictors are dated at the close of the origin day t . Rolling estimates use the 750 trading days ending on day t . The last specification lets the persistence state change the weights on the HAR components, the channel through which slower decay of shocks would act.

4.2. Estimation

Each model is re-estimated by ordinary least squares at every weekly origin on an expanding window of that stock's past observations. Because targets overlap for $h > 1$, a past observation enters the regression only if its target window has closed by the origin: the observation for an earlier origin s is used at origin t only if its target window, days $s + 1$ to $s + h$, ends no later than day t . Without this restriction the last observations of each training sample contain variance that has not yet been realized; at $h = 22$ this inflated the apparent gain of the richer models in an earlier version of this study. The first 431 origins (up to 14 June 2013) form the initial training sample, and all models are evaluated over the same 646 origins, from 17 June 2013 to 14 April 2026, on the same (date, stock) cells.

4.3. Evaluation

The loss function is the squared error of the log-variance forecast, pooled over stocks and dates; Appendix B reports the QLIKE loss for variance forecasts corrected point-in-time for the log transformation. Comparisons are based on the cross-sectional mean loss differential per date, $\bar{\delta}_t = N^{-1} \sum_i (L_{i,t}^{(1)} - L_{i,t}^{(2)})$, which absorbs the strong cross-sectional correlation of forecast errors. We report three tests on $\bar{\delta}_t$, each with a Newey–West variance and bandwidth $\max(\lceil h/5 \rceil - 1, \lfloor 4(T/100)^{2/9} \rfloor)$, which is six weeks for $T = 646$:

- the Diebold–Mariano test of equal mean loss (Diebold & Mariano, 1995) with the small-sample correction of Harvey et al. (1997);
- for linear models that nest HAR-X, the Clark and West (2007) test, which adjusts the loss differential for the noise introduced by estimating parameters whose population value is zero, and so tests whether the extra predictors have nonzero population coefficients;
- the conditional predictive ability test of Giacomini and White (2006), with the constant and the most recent observed differential $\bar{\delta}_{t-\lceil h/5 \rceil}$ as instruments, which asks whether the estimated model, with its estimation error, forecasts better.

4.4. Registration

The study was conducted in stages. Before the results of each stage were computed, its specifications, comparisons, tests and decision rules were recorded in a dated experiment ledger. Under the rule for the registered HAR-X extensions, a specification adds information beyond HAR-X at a given horizon only if its Holm-adjusted (Holm, 1979) Clark–West p -value is below 0.05 and its mean squared error is lower than that of HAR-X. After the results were known, we added the requirement that the gain be at least 0.5%; we report it separately because it was not registered. A complete audit of the pipeline preceded the final run. Every correction was registered before the corrected run and is listed in the ledger, which is published with the code (Appendix D).

5. Results

5.1. Implied volatility accounts for the gains over HAR

Table 4 compares each model with HAR. Adding the VIX and MOVE lowers the pooled mean squared error by 3.1% at one day and 5.5% at one week (Diebold–Mariano statistics of 5.23 and 5.35), and by 1.8% at one month, which is not significant. The gains are broad: HAR-X improves on HAR for 111 of the 115 stocks at the weekly horizon.

Table 4. Out-of-sample accuracy relative to HAR.

Model	$h = 1$		$h = 5$		$h = 22$	
	Δ MSE	DM	Δ MSE	DM	Δ MSE	DM
HAR	0.6146		0.3381		0.2632	
HAR-X	+3.11%	+5.23***	+5.54%	+5.35***	+1.77%	+0.95
HAR + own persistence	-0.61%	-3.77***	-0.09%	-0.24	-1.88%	-3.21***
HAR + cross-sectional state	-0.23%	-0.95	+0.00%	+0.01	-1.61%	-0.68
HAR + sector state	-0.09%	-1.71*	+0.13%	+0.70	-0.01%	-0.03
HAR + d , VIX, MOVE, $d \times$ VIX, $d \times$ MOVE	+2.89%	+4.02***	+5.58%	+4.47***	+0.41%	+0.21
Full model C	+2.14%	+2.58***	+5.79%	+4.22***	-1.54%	-0.64

Notes: Pooled out-of-sample MSE of log mean future Parkinson variance, 646 weekly origins (17 June 2013 to 14 April 2026) $\times$ 115 stocks, common cells. The HAR row reports its MSE; other rows the percentage reduction relative to HAR (positive is better) and the panel Harvey–Leybourne–Newbold Diebold–Mariano statistic. *, **, ***: 10%, 5%, 1%.

Added to HAR without implied volatility, the persistence blocks do not improve forecasts. Own-stock persistence worsens them significantly at the daily and monthly horizons, and the effects of the cross-sectional and sector states are not significant at the 5% level. The only specifications that improve on HAR, A_5 and C, are those that contain the VIX and MOVE. Their gains are close to those of HAR-X at the weekly horizon (5.6% and 5.8%, against 5.5%) and smaller at the daily horizon (2.9% and 2.1%, against 3.1%); at one month model C is worse than HAR. The stock-level results point the same way: model C improves on HAR for 107 stocks at the weekly horizon but for only 48 at the monthly horizon.

5.2. Persistence beyond HAR-X

Table 5 compares the persistence specifications directly with HAR-X. The three registered extensions change the mean squared error by between -2.0% and $+0.003\%$. None has a significant Diebold–Mariano statistic in its favor, and at one month one is worse at the 5% level and another at the 10% level. Model C gains 0.26% at one week ($t = 0.41$) and is significantly worse at one day (-1.0% , $t = -2.14$) and at one month (-3.4% , $t = -2.69$).

Table 5. Incremental accuracy of the persistence specifications relative to HAR-X.

Specification	h	ΔMSE	DM	CW	CW p (Holm)	GW p
HAR-X + cross-sectional state [†]	1	-0.03%	-0.10	+3.76***	0.001	0.743
	5	+0.00%	+0.01	+4.44***	0.000	0.923
	22	-2.01%	-1.77*	+1.11	0.403	0.133
HAR-X + sector state [†]	1	-0.10%	-0.35	+2.77***	0.017	0.668
	5	-0.13%	-0.34	+2.49***	0.032	0.543
	22	-0.86%	-1.30	+0.94	0.403	0.418
HAR-X + state $\times$ HAR terms [†]	1	-0.37%	-0.93	+3.09***	0.007	0.401
	5	-0.39%	-0.66	+2.44***	0.032	0.282
	22	-1.77%	-2.04**	+0.61	0.403	0.108
Full model C	1	-1.00%	-2.14**	+5.96***	0.000	0.064
	5	+0.26%	+0.41	+8.31***	0.000	0.697
	22	-3.37%	-2.69***	+6.63***	0.000	0.018

Notes: ΔMSE : percentage reduction in pooled MSE relative to HAR-X (positive favors the specification). DM: panel HLN Diebold–Mariano statistic (two-sided). CW: Clark–West statistic for the nested comparison (one-sided). Holm adjusts the nine tests of the three registered specifications ([†]) jointly and the three tests of model C jointly. GW: Giacomini–White conditional predictive ability test with instruments $(1, \delta_{t-k})$, $k = \lceil h/5 \rceil$. All statistics use the cross-sectional mean loss differential per date and a Newey–West bandwidth of $\max(\lceil h/5 \rceil - 1, \lfloor 4(T/100)^{2/9} \rfloor) = 6$.

The Clark–West statistics appear to point the other way. They are large and positive for every specification at one day and one week, and the Holm-adjusted p -values are below 0.05 for all three registered extensions at those horizons. Two specifications therefore formally meet the registered rule: the cross-sectional extension at one week, with a gain of 0.003%, and model C at one week, with a gain of 0.26%. Neither gain is economically meaningful, and neither meets the 0.5% threshold that we added after seeing the results.

The two tests answer different questions, and the Giacomini–White test resolves the apparent conflict. The Clark–West test asks whether the persistence coefficients are zero in population; its adjustment removes the noise that estimating these coefficients adds to the forecasts of the larger model. A rejection therefore indicates that the persistence terms have predictive content once HAR-X is controlled for. The Giacomini–White test asks whether the estimated model forecasts better, conditional on recent relative performance. It never favors a persistence specification: its p -values for the registered extensions lie between 0.11 and 0.92, and its only rejection, for model C at one month ($p = 0.018$), goes against the persistence model, as does the near-rejection at one day ($p = 0.064$). The predictive content of the persistence terms is genuine but small, and estimating the additional coefficients stock by stock (11 of them for model C) costs more accuracy than it adds.

Figure 2 shows when the gains and losses accrue. The gain of HAR-X over HAR accumulates steadily at the daily and weekly horizons, with jumps in March 2020. The gain of model C over HAR-X declines steadily at the daily and monthly horizons and stays close to zero at the weekly horizon, where its small positive balance appears only in 2025.

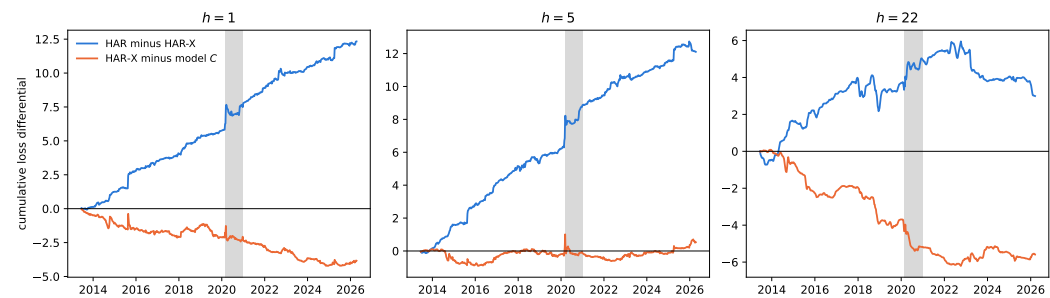


Figure 2. Cumulative sum over forecast origins of the cross-sectional mean squared-error differential. Blue: HAR minus HAR-X (rising when implied volatility improves forecasts). Orange: HAR-X minus model C (rising when persistence improves forecasts beyond implied volatility). Common cells; shading marks March to December 2020.

5.3. Information timing

The comparison between HAR and HAR-X is sensitive to a detail that is easily overlooked. If the HAR terms are built from variance through day $t - 1$, for example by lagging all regressors by one day to avoid look-ahead bias, while the VIX is the close of day t , the implied-volatility model receives one more day of information than the benchmark. Table 6 re-estimates both models with HAR inputs ending at $t - 1$ and compares them with the aligned specifications used elsewhere in the paper.

Table 6. Information timing and the apparent value of implied volatility.

HAR inputs	h	MSE HAR	MSE HAR-X	HAR-X gain (DM)
Through $t - 1$	1	0.6691	0.6350	+5.11% (+4.56)
	5	0.3637	0.3354	+7.76% (+4.62)
	22	0.2712	0.2633	+2.90% (+1.46)
Through t	1	0.6146	0.5955	+3.11% (+5.23)
	5	0.3381	0.3193	+5.54% (+5.35)
	22	0.2632	0.2586	+1.77% (+0.95)
HAR, t vs $t - 1$	1			+8.15% (+6.39)
	5			+7.04% (+5.00)
	22			+2.93% (+3.41)

Notes: VIX and MOVE are closes of the origin day t in every row. Through $t - 1$: HAR terms built from variance through the day before the origin and the previous day's return. Through t : the aligned specification used everywhere else. Common cells of all four forecast panels; DM statistics in parentheses.

Aligning the inputs lowers the error of HAR by 8.1% at one day, 7.0% at one week and 2.9% at one month, all significantly. Because HAR-X already contains the VIX of day t , it gains less from the alignment, and its advantage over HAR falls from 5.1% to 3.1%, from 7.8% to 5.5% and from 2.9% to 1.8%. Between 29% and 39% of the apparent value of implied volatility is therefore a timing artifact. Earlier versions of this study used the misaligned convention and reported the larger gains. Studies that compare HAR with implied-volatility regressors should verify the same alignment.

5.4. Pooled, term-structure and market-level designs

The per-stock design invites three objections, and Table 7 addresses each with a design registered before its results were computed.

Table 7. Pooled, term-structure and market-level designs.

Design	Comparison	h	Δ MSE	DM	CW p (Holm)
Pooled, fixed effects	Full model vs HAR-X	1	+0.40%	+1.19	0.000
		5	+0.34%	+0.79	0.000
		22	-1.36%	-1.30	0.627
	HAR-X + cross-sectional state vs HAR-X	1	+0.34%	+1.18	0.004
		5	+0.19%	+0.53	0.016
		22	-1.00%	-0.96	0.627
Duration target	Full model vs HAR-X	22	-1.81%	-3.92***	0.026
	HAR-X + cross-sectional state vs HAR-X	22	-0.35%	-1.42	0.677
Market level	Full market model vs market HAR-X	1	+3.42%	+1.77*	0.000
		5	+2.01%	+1.45	0.000
		22	-1.93%	-1.00	0.270

Notes: Pooled: one regression across stocks with stock fixed effects (within transformation on training means), refit at every origin. Duration target: log ratio of mean future variance over 22 and 5 days. Market level: one series, the cross-sectional mean of the stock targets, with market HAR terms, VIX, MOVE and the persistence state. Each design was specified before its results were seen.

First, per-stock regressions estimate many coefficients from 430 to about 1070 weekly observations, while the persistence state is common to all stocks. A pooled regression with stock fixed effects estimates each coefficient once from the whole panel. Pooling helps: model C now improves on pooled HAR-X by 0.40% at one day and 0.34% at one week, and the Clark–West rejections survive the Holm adjustment. The gains are, however, below the 0.5% threshold and not significant by the Diebold–Mariano test, and at one month the sign reverses.

Second, persistence might describe how long volatility stays high rather than how high it is. We therefore forecast the slope of the variance term structure, defined as the log ratio of mean variance over the next 22 days to mean variance over the next 5. HAR improves on an expanding-mean benchmark for this target by 3.3%, so the slope is forecastable, but adding persistence to HAR-X makes the forecasts worse, significantly so for model C (-1.8% , $t = -3.92$).

Third, the state is a market-wide quantity and may matter only at the market level. For a single series, the cross-sectional mean of the stock targets, the persistence state improves on a market HAR-X by 3.4% at one day and 2.0% at one week, with Diebold–Mariano statistics of 1.77 and 1.45. These are the largest gains in the study, but with one series and 646 origins they are not significant. An exploratory analysis in Appendix C suggests that they reflect the slowly decaying variance risk premium after crises rather than persistence.

5.5. Calm and stressed periods

Splitting the evaluation period by the quartiles of the VIX on the forecast dates reveals no regime in which persistence improves forecasts significantly. In the highest VIX quartile model C improves on HAR-X by 2.4 percentage points at one week, with a Diebold–Mariano statistic of 1.36; in the lowest quartile it is worse at every horizon; from March to December 2020 it is 0.2 points worse at one week and 5.6 points worse at one month. With 162 dates per quartile and 43 in 2020, these comparisons are descriptive.

6. Robustness and Economic Value

6.1. Estimators, windows, targets and learners

Panel A of Table 8 varies the construction of the persistence predictors and the target at the weekly horizon, where model C performs best. Its gain over HAR remains between 5.2% and 6.4% when d is estimated by local Whittle instead of GPH, over 500 or 1000 days instead of 750, or for the more or less liquid half of the sample, and it is 2.6% when the

target is the mean squared return. Each gain is of the same order as that of HAR-X in the baseline, so none of these choices alters the conclusions of Section 5. A GARCH(1,1) forecast, corrected point-in-time for its level, is 19.9% worse than HAR, consistent with the HAR family being the stronger benchmark for range-based variance.

Table 8. Robustness of model C at $h = 5$, and the machine-learning estimators.

<i>Panel A. Model C versus HAR at $h = 5$</i>		
Variant	Δ MSE vs HAR	DM
Headline (GPH, $W = 750$)	+5.79%	+4.22***
Local Whittle instead of GPH	+5.98%	+4.13***
Window 500 days	+5.33%	+3.56***
Window 1000 days	+6.19%	+4.16***
Target: squared returns	+2.63%	+2.89***
Less liquid half	+6.36%	+4.16***
More liquid half	+5.19%	+3.98***
GARCH(1,1) instead of model C	-19.94%	-9.04***
<i>Panel B. Model D: model C's predictors, other estimators (ΔMSE vs HAR-X at $h = 1/5/22$)</i>		
Estimator	Δ MSE	DM
Lasso	-0.8% / -1.3% / -3.1%	-2.6 / -2.2 / -2.6
Ridge	-1.5% / -1.9% / -4.2%	-3.7 / -2.6 / -3.0
Elastic net	-0.8% / -1.4% / -3.3%	-2.5 / -2.2 / -2.5
Random forest	-5.8% / -6.2% / -7.0%	-4.5 / -3.9 / -3.2
Gradient boosting	-6.2% / -6.1% / -10.3%	-6.2 / -5.4 / -4.2

Notes: Panel A: percentage MSE reduction of model C relative to HAR at $h = 5$ under each variant; the GARCH row compares a level-corrected GARCH(1,1) forecast with HAR. Panel B: shrinkage hyperparameters and gradient-boosting tree size, leaf size and number of trees chosen by time-series cross-validation inside each training window; the random forest uses 200 trees with a minimum leaf of 20; refit every 20 weeks.

Panel B replaces the linear regression on the 18 predictors of model C by five machine-learning estimators: the lasso, ridge regression and the elastic net, whose penalties are chosen by time-series cross-validation within each training window; a random forest; and gradient boosting, whose tree size, minimum leaf size and number of trees are chosen in the same way. All five are significantly worse than HAR-X at every horizon. The shrinkage estimators lose 0.8% to 4.2%, and the tree ensembles 5.8% to 10.3%. In this sample, neither regularization nor nonlinearity extracts information from the persistence predictors that the linear HAR-X model misses. Tuning matters for this conclusion: gradient boosting with fixed default settings (400 trees with 31 leaves) loses 12% to 21% relative to HAR, an artifact of overfitting 430 to about 1070 training observations that tuning largely removes.

6.2. Volatility-managed portfolios

A forecast that is statistically no more accurate can still be economically useful. We examine this with the volatility management of [Moreira and Muir \(2017\)](#), applied stock by stock: each week the position in stock i is scaled by $c_{i,t}/\hat{\sigma}_{i,t+5|t}^2$, where $\hat{\sigma}_{i,t+5|t}^2$ is the weekly-horizon variance forecast of a given model and $c_{i,t}$ is set in real time so that the managed and unmanaged positions have the same past variance. Positions are funded at the three-month Treasury bill rate, and returns are measured in excess of it. In addition to the HAR, HAR-X and model C forecasts, we include a model-free rule that scales positions by trailing 22-day variance.

Table 9. Volatility-managed portfolios (excess returns, weekly, 2014–2026).

Portfolio	Sharpe	Sharpe, 10 bp	Turnover	Sharpe, COVID	Max drawdown
Equal weight	0.76	0.76	1.1	0.88	-30.6%
Trailing 22-day variance	0.76	0.69	6.6	1.09	-15.7%
HAR	0.75	0.64	11.5	1.29	-17.3%
HAR-X	0.76	0.66	10.5	1.38	-15.6%
Model C	0.76	0.66	10.3	1.62	-15.8%

Notes: Each stock's position is scaled by $c_{i,t}/\hat{\sigma}_{i,t+5|t}^2$ with $c_{i,t}$ set in real time so that the managed and unmanaged positions have equal past variance; positions are funded at the three-month Treasury bill rate; 594 weeks after a 52-week warm-up. Turnover: notional traded per unit of capital per year, including drift. COVID: March to December 2020 (43 weeks). Sharpe differences, Ledoit–Wolf HAC p (block bootstrap p): model C vs HAR-X 0.94 (0.93) full sample, 0.10 (0.31) COVID; model C vs equal weight 0.97 (0.98).

Table 9 shows no economic value from any of the forecasts. Every managed portfolio has a full-sample Sharpe ratio of 0.75 or 0.76, equal to that of the equal-weighted portfolio, and none of the differences is close to significant. Management lowers volatility and roughly halves the maximum drawdown, but it reduces expected excess returns in proportion. The managed portfolios trade about ten times their capital each year, and at a cost of 10 basis points per unit traded their Sharpe ratios fall to 0.64–0.66, below that of equal weighting. The one period in which model C stands out, March to December 2020, contains 43 weekly observations, and its advantage over HAR-X in that period (a Sharpe ratio of 1.62 against 1.38) is not significant. These results are in line with the evidence that volatility management of factor portfolios is fragile out of sample (Barroso & Detzel, 2021; Cederburg et al., 2020).

7. Discussion

7.1. Why persistence does not improve forecasts

Section 3 shows that the rolling persistence state is largely a smoothed record of the largest volatility episodes of the preceding two to four years. It rises when such an episode enters the window and falls, on a date fixed by the window length, when the episode leaves. Sections 5 and 6 show that, once the VIX and MOVE are in the model, the state does not significantly improve forecasts in any of the registered designs. We interpret the first result as the explanation for the second. Implied volatility is a forward-looking price that already reflects recent turmoil, partly through a variance risk premium that rises after crises and decays slowly (Andersen et al., 2015; Bates, 2000; Bekaert & Hoerova, 2014). A backward-looking indicator of when the last crisis occurred has little to add. The Clark–West rejections show that it adds some information in population; the Giacomini–White tests show that this contribution is too small to survive estimation.

These results do not imply that volatility lacks long memory, or that the persistence of volatility shocks never changes. The full-sample estimates in Table 1 are large, and our short-memory null reproduces their calm-period level, consistent with a long literature showing that level shifts and long memory are difficult to separate (Diebold & Inoue, 2001; Perron & Qu, 2010; Varneskov & Perron, 2017). Our claim is narrower: the time variation in rolling estimates is not evidence of time variation in persistence. It is the pattern that a process with constant parameters produces when a few large episodes pass through a fixed window.

7.2. Implications for applied work

Three practices follow for research that uses rolling memory estimates, Hurst exponents or similar windowed statistics. First, results should be reported for several window lengths, with a check of whether the turning points move with the window; a structural change should not. Second, the windowed statistic can be regressed on the largest value of

an observable stress measure inside the same window; a high R^2 indicates that the statistic reflects window composition rather than dynamics. Third, the series can be compared with a short-memory null into which the observed episodes are injected, and with estimators and tests designed to separate long memory from level shifts (Hou & Perron, 2014; Qu, 2011). These are standard tools in fields that encountered the same problem earlier, from hydrology (Klemeš, 1974) to early-warning indicators in ecology and climate science (Boettiger & Hastings, 2012; Dakos et al., 2008; Ditlevsen & Johnsen, 2010) and sliding-window connectivity in neuroscience (Leonardi & Van De Ville, 2015).

Three lessons for forecasting studies extend beyond this application. The benchmark and the challenger must use the same information set, down to the day: a one-day lag in the inputs of the benchmark inflated the value of implied volatility by about a third. When a nested comparison yields a Clark–West rejection without a lower loss, a Giacomini–White test indicates whether the estimated model is useful. Finally, machine-learning comparisons are informative only if every learner is tuned under the same protocol: with default settings gradient boosting lost 12–21% relative to HAR, against 0.2–8% once tuned.

7.3. Limitations

The panel consists of current S&P 500 members with long histories. Firms that left the index are missing, which removes some of the most volatile histories; this omission affects the level of volatility and of the memory estimates more than the window mechanism, which is a property of the estimator. The variance proxy is the daily range rather than intraday realized variance. Its sampling noise lowers the level of $\hat{d}$ and of the Hurst exponent but does not change the dates at which episodes enter and leave the window. Returns are price returns, which matters only for the portfolio exercise. The forecast evaluation period, 2013 to 2026, contains one extreme episode, in 2020, and several moderate ones; the window analysis also uses 2008. Markets without a liquid implied-volatility index are where persistence could add the most, but in such markets the largest recent realized variance is directly available and would be the natural benchmark. Finally, the evidence on the window mechanism concerns memory in volatility; a general analysis across estimators, asset classes and markets is left for future work.

8. Conclusions

Rolling long-memory estimates of stock volatility rise sharply in crises, and it is tempting to read them as a measure of how persistent the market has become. For 115 S&P 500 stocks over 2001–2026 they do not improve volatility forecasts beyond implied volatility under any design we registered: per-stock and pooled regressions, the level and the term structure of future variance, a market-level series, linear models and tuned machine-learning estimators, calm and stressed periods, and volatility-managed portfolios. The evidence points to a simple reason: the rolling persistence state records which extreme episodes lie inside its window. It jumps when an episode enters and falls when the episode leaves, at dates that move one-for-one with the window length; a short-memory process reproduces this pattern; and 83% of its variation is explained by the largest VIX inside the window. Researchers who use rolling memory estimates as a state variable should check for these signatures before interpreting them. Forecasting studies that compare HAR with implied volatility should align the information sets of the two models, which here removes a third of the apparent value of the VIX.

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Appendix A. Roughness at the daily frequency

The increment-scaling estimate of the Hurst exponent of daily log Parkinson variance averages 0.060 across stocks (10th–90th percentile 0.053–0.069). Read literally, this indicates volatility much rougher than Brownian motion, for which the exponent is 0.5. We ask whether a process without fractional roughness, observed through the Parkinson estimator, produces the same value. Latent log variance is the sum of two independent AR(1) components, a Markovian model. The observed series adds the sampling error of the range estimator, $\eta_t = \ln(R_t^2/4 \ln 2)$ minus its mean, where R_t is the range of a standard Brownian motion over one day sampled at 390 one-minute steps; its variance, 0.36, is fixed by theory and is about a third of the variance of observed log variance. The two autoregressive coefficients and the variance split are fitted to the median autocorrelations of observed log variance at nine lags from 1 to 250 days, which gives a fast component with coefficient 0.66 and a slow one with coefficient 0.995. Over 200 simulated paths of 6136 days the same estimator gives 0.064 (standard deviation 0.004) on the observed series, matching the data, and 0.19 on the noise-free latent series. Two features of daily data therefore produce the low estimate without any roughness: the sampling noise of the range, which lowers the estimate from 0.19 to 0.06, and a fast mean-reverting component, which already holds it well below 0.5 at lags of up to 21 days. Daily range data cannot identify roughness, in line with Cont and Das (2024). The code is module 16 of the repository.

Appendix B. Additional forecast results

QLIKE. Variance forecasts are obtained from the log forecasts with a point-in-time smearing correction: the mean of the exponentiated past out-of-sample errors whose targets had been observed by the origin, with at least 26 such errors. Against HAR-X, no specification has a significantly lower QLIKE loss at the 5% level; the largest Diebold–Mariano statistic in favor of persistence is 1.67 (cross-sectional extension, one day), and model C is significantly worse at one month (−2.08).

Machine learning. The lasso, ridge and elastic-net penalties are chosen by five-fold expanding time-series cross-validation with a gap of $\lceil h/5 \rceil$ rows between training and validation folds. Gradient boosting (LightGBM, learning rate 0.05) chooses the number of leaves $\{2, 4, 15\}$, the minimum leaf size $\{20, 50\}$ and the number of trees $\{25, \dots, 800\}$ by three-fold time-series cross-validation in each training window. The random forest uses 200 trees with a minimum leaf of 20. Each learner is refit every 20 weeks on the embargoed training rows. With fixed defaults (400 trees, 31 leaves) gradient boosting loses 15.9%, 11.9% and 21.2% to HAR at the three horizons; tuned, it loses 2.9%, 0.2% and 8.3%.

Appendix C. The market-level gain

On the market-level series the persistence state improves on market HAR-X by 3.4% and 2.0% at one day and one week (Table 7). An exploratory analysis, specified after that result and therefore not a formal test, asks what accounts for the gain. In a specifi-

cation in levels, the interaction of the VIX with the logarithm of its maximum over the previous three years has a negative coefficient ($t = -2.7$ at both horizons): the slope of future realized variance on the VIX is about 40% lower when a crisis is recent. In logs the interaction disappears ($t = 0.33$ and 0.26), and the three-year maximum enters only as a level shift, which is absorbed by the trailing 250-day gap between VIX-implied and realized variance. That gap alone improves on the log market HAR-X model by 5.0% to 6.3% (Diebold–Mariano statistics of 2.7 to 3.5). The market-level gain is therefore best interpreted as the slowly decaying variance risk premium after crises (Andersen et al., 2015; Bates, 2000), for which the persistence state may act as a proxy because both depend on how recently the last crisis occurred. A test of this interpretation requires data not used here, such as earlier index history or other markets.

Appendix D. Registration record and audit

The study was carried out in stages. Before each stage the specifications, comparisons, tests and decision rules were written into an experiment ledger kept under version control; results were added afterwards and earlier entries were not edited. The ledger, the code and a list of every correction are in the repository named in the Data Availability Statement. The main stages were:

- an embargo on training observations whose targets were not yet realized at the origin, which removed an apparent gain of the persistence model at the monthly horizon;
- three HAR-X extensions with a fixed decision rule and Holm adjustment;
- pooled, term-structure and market-level designs, point-in-time winsorization and real-time portfolio normalization;
- a full audit of the pipeline, whose corrections were registered before the final run: aligned information timing (Table 6), missing rather than floored zero-range days, memory estimated on log variance throughout, a Newey–West bandwidth that grows with the sample, point-in-time QLIKE, time-series cross-validation for every learner, a level-corrected GARCH benchmark, common evaluation cells and dates, and portfolio returns in excess of the bill rate;
- the window-inclusion analysis (Table 2), the Giacomini–White tests and the timing comparison, each registered with a prediction before it was run.

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